AI 500 - Foundations of AI

QUIZ 6 — Answer Key

November 15, 2025

Q1. Linear model and ATE

The model is:

$$Y_i = \beta X_i + \gamma T_i + \epsilon_i$$

Given: $\gamma = 5$

ATE Explanation: In a linear regression with binary treatment T, the coefficient γ represents the average causal effect of treatment (assuming ignorability and no omitted confounding).

$$\text{ATE} = \gamma = 5$$

Reasoning: The treatment increases Y by 5 units on average compared to no treatment.

Q2. Multivariate regression and unobserved confounding

Answer: False

Reasoning: Multivariate regression can only adjust for *observed* covariates. Unobserved confounders that affect both treatment and outcome cannot be controlled and therefore bias remains.

Q3. Drug study with gender confounding

Treatment	Recovery Rate	N
Drug	80%	50
No Drug	85%	50

Observation: The drug group has more women (70%), the no-drug group only 30%. Women have lower recovery rates overall.

Answer: This does not represent a causal effect.

Reasoning: The observed difference (80% vs. 85%) may be due to confounding by gender, not the drug. To estimate the true causal effect, we must adjust for gender (e.g., stratification or regression).

Q4. Tutoring program CATE

Student	Prior Performance	T	Post-test Y
1	High	1	90
2	High	0	85
3	Low	1	70
4	Low	0	60

CATE calculations:

High performers:

$$90 - 85 = 5$$

Low performers:

$$70 - 60 = 10$$

Answer: The tutoring program is more effective for **low-performing students** (10-point gain vs. 5-point gain).

Q5. Teaching method paradox

Department	New Method	Old Method	N
Math	90%	80%	50
History	70%	60%	50
Science	75%	85%	50
Combined	78%	82%	150

Observation:

- Math and History: New > Old
- Science: Old > New

Overall: Old appears better (82% > 78%).

Answer: No paradox (not Simpson's Paradox).

Reasoning: Simpson's Paradox requires *all* subgroups to show the same direction of effect, which reverses when aggregated. Here, Science already favors the old method. Thus the reversal is due to aggregation effects, not Simpson's Paradox.